



Faculty of Information Technology

Semester 2, 2026

**FIT5145: Foundations of Data Science**

**Assignments 1 & 3: Business and Data Case Study**

1. **Assignment 1 (Proposal):** Draft a proposal to introduce a data science project of interest. The due date of Assignment 1 is: **Friday, 28 August 2026, 11:55 PM (Week 5).**
  2. **Assignment 3 (Written Report + Recorded Presentation + Interview):** Write a comprehensive report on your data science project and prepare a 4-minute recorded presentation on your project. The due date of Assignment 3 (the final project report and recorded presentation) is: **Friday, 16 October 2026, 11:55 PM (Week 11)** and the interview will be held **in the Week 12 applied class.**
- Both Assignment 1 and 3 are **individual** assignments.
  - **Please do NOT zip your submission files.** Zip file submission will have a penalty of 20% of the total mark of the assignment. Failure to submit any of the required files will result in a loss of marks associated with the evaluation of those files. If you submit an incorrect version of a required file, you may request a resubmission; however, a late submission penalty will apply to the entire assignment to ensure marking fairness and consistency. **Please ensure that you submit the correct versions of all required files!**

### **Focus of the Project Proposal**

Assignments 1 and 3 require you to develop a **novel** data science project proposal that introduces an original approach to solving a significant real-world problem using data science methods. You are expected to go beyond existing studies by identifying unique problem statements, proposing innovative methodologies, or applying data science techniques in new contexts. Your proposal should demonstrate your ability to define a novel and important problem, identify relevant datasets, select appropriate methodologies, and develop effective evaluation strategies. You can choose any of the proposed business scenarios, such as agriculture, education, finance, the gaming industry, healthcare, social media, and sports, or any other scenario where data science can be used to solve a real-world problem.

### **Assignment 1: Proposal (15%)**

**Weight:** 15% of the unit mark

**Submission format:** one PDF file

**Size:** up to 1000 words.

**What you need to do:**

- Choose a data science project.

- Write the initial three sections: (1) *Introduction*; (2) *Related Work*; and (3) *Business Model* (*References* as well to support your project) of the report, as detailed in the specification of *Assignment 3: Written Report + Recorded Presentation + Interview* below.

**Important:** If you use GenAI for this assignment, please ensure the following content is included on the FIRST page of your Assignment 1 submission:

- What type of GenAI (e.g., ChatGPT, Gemini, DeepSeek) did you use for this assignment?
- What did you use GenAI for in this assignment? Potential answers include:
  - Brainstorming and Idea Generation
  - Literature Exploration Support
  - Dataset Discovery
  - Methodology Selection Support (e.g., Machine Learning)
  - Writing Structuring and Clarity Improvement
  - Others (please specify)

Your answers to these questions will help guide how GenAI can be incorporated to enhance teaching and learning in this unit in future semesters.

### **Assignment 3: Written Report (15%) + Recorded Presentation (5%) + Interview and Responses to Questions (10%)**

#### **1. Assignment 3: Written Report**

**Weight:** 15% of the unit mark

**Submission format:** one PDF file and one RMD file (for demonstration in the Characterising and Analysing Data section)

**Size:** up to 2500 words

This report is your *comprehensive* analysis of how data science can be used to help solve a significant real-world problem. Please answer the following question in the **FIRST** page of your Assignment 3 submission:

- Have you selected a topic for Assignment 3 that is different from the one that you used for Assignment 1 (i.e., have you rewrote the first three sections of the report)?

**Your report should have the following sections:**

##### **1. Introduction**

- Clear articulation of the specific problem the project aims to solve.
- Background and context of the problem.
- Importance of the problem (why it matters).
- Specific goals of the project.

##### **2. Related Work**

- Summary of existing research, projects, or industry solutions related to the problem.
- Identification of gaps in current approaches.
- Why or how your project should be considered as *novel*.

### 3. Business Model

- What are the project's data business model and level of analytics?
- What kind of benefits or values the project can create for the specific business area?
- Who are the primary stakeholders and how will they benefit from the project?

### 4. Characterising and Analysing Data:

- Discuss potential data sources and analyse their characteristics (e.g., the 4 V's), evaluate the required platforms, software, and tools for data processing and storage based on the specific characteristics of the data and consider potential options (e.g., platforms, software, and tools) if your project expands in the future.
- Specify the data analysis approach, including the overall process and the technical and statistical methods (e.g., machine learning techniques) applicable to the project. Provide a rationale for the selected approach and methods, and discuss the expected outcomes.
- Demonstration: identify a usable dataset for the proposed project and perform some analysis on the identified dataset to demonstrate the feasibility of the project, using R (e.g., extracting insights from the dataset and building machine learning models), and report the analysis process and result in the demonstration section of a final report. *Note: Please include a link to download the dataset in the final report, and upload the R markdown file created for data analysis on Moodle.*
- *Note: The specification of the Data Analysis Approach section needs to be different from the Demonstration section and should be described separately. In the Data Analysis section, provide a high-level and comprehensive overview of your overall analysis approach. In the Demonstration section, apply part of your proposed analysis approach to a subset of your dataset to demonstrate the feasibility of your project.*

### 5. Standard for Data Science Process, Data Governance and Management

- Describe any standards used in your data science process
- Describe any practices for data governance and management in the project, e.g., how to address key issues such as data accessibility, security, and confidentiality, as well as potential ethical concerns related to data usage.

The sections would present aspects of Weeks 1-10 of the unit for your chosen case study.

The **maximum word limit** for the report (Assignment 3) is **2500** words. It may include some/all of your Assignment 1, modified if needed (counted in the 2500 word total). References at the end of the report (i.e., URLs and academic publications) are not included in the word count. Note that staying within the word limit demonstrates your ability to write concisely.

## 2. Assignment 3: Recorded Presentation (5%) + Interview (10%)

### (1) Recorded Presentation (5%)

**Submission format:** One video file in the .mp4 format (submitted through Moodle)

**Length:** Approximately 4 minutes

You are required to prepare and submit a recorded presentation introducing your data science project. You are required to record a 4-minute presentation by capturing your slides on screen while recording your voice and face simultaneously using a microphone and camera. The recording should clearly explain the key sections covered in your report, including the motivation for the project, the proposed solution, the

data sources, the planned analytical methods, the expected outcomes, the significance of the project, and other relevant aspects of your project.

The recorded presentation must be submitted via Moodle before your allocated Week 12 applied class. The teaching team will watch your recording before conducting the interview.

## **(2) Interview and Responses to Questions (10%)**

During your allocated Week 12 applied class, your tutor will conduct an individual interview of approximately **4 minutes**, during which you will be asked **2–3 questions** about your project. The questions may focus on your project motivation, methodology, data sources, analytical techniques, project feasibility, expected outcomes, and other aspects of your report and recorded presentation.

The purpose of the interview is to assess your understanding of your project and your ability to explain and justify your proposed data science solution.

## **How you will be assessed**

See the marking rubric to understand how we will grade your assignments.

**Assignment 1 proposal:** The 15% awarded for your proposal is broken down into the following categories:

- **Problem Clarity (2%):** Is the problem well-articulated and clearly defined? Is the goal of the project clearly stated?
- **Problem Importance (4%):** Does the project have real-world applications? Does it address key social, environmental, or business challenges and demonstrate potential for significant social impact?
- **Business Model Analysis (4%):** Is the data business model of the project clearly articulated? Is the level of analytics provided by the project clearly identified, such as descriptive, predictive, or prescriptive analytics? Are the benefits, value and challenges of the project clearly outlined? Are the primary stakeholders identified and their needs addressed?
- **Novelty (5%):** Does the project address an important and novel problem? Does it introduce a new or unconventional approach? Does it tackle an underexplored or emerging issue in data science?

**Assignment 3 report:** You will be assessed on your ability to:

- define the problem, provide background and significance, outline specific goals, analyse the business domain and its value creation, identify key stakeholders and their benefits, summarize existing research or industry solutions, highlight gaps in current approaches, and justify the project's novelty and potential impact (You can reuse the content from Assignment 1 for this section);
- discuss potential data sources and analyse their characteristics (e.g., the 4 V's) and evaluate the required platforms, software, and tools for data processing and storage based on the specific characteristics of the data and consider potential options (e.g., platforms, software, and tools) if your project expands in the future;
- specify the data analysis approach, including the overall process and the technical and statistical methods (e.g., machine learning techniques) applicable to the project. Provide a rationale for the selected approach and methods, and discuss the expected outcomes;

- identify a usable dataset for the proposed project and perform some analysis on the identified dataset to demonstrate the feasibility of the project, using R (e.g., extracting insights from the dataset and building machine learning models), and report the analysis process and result in the demonstration section of a final report;
- describe any standards used in your data science process and practices for data governance and management in the project, e.g., how to address key issues such as data accessibility, security, and confidentiality, as well as potential ethical concerns related to data usage;
- think critically and creatively, providing justification and analysis;
- Provide a high-quality report in terms of structure, organisation, formatting, clarity of writing, grammar, spelling, referencing, and adherence to the word limit.

For both assignments, make sure that any resources you use are acknowledged in your report. You may need to review the [FIT citation style](#) to make yourself familiar with appropriate citing and referencing for this assessment. Also, review the [demystifying citing and referencing](#) guide for help.

Please also make sure that the Turnitin scores will be generated properly for your submissions. If a submission receives a high Turnitin score (e.g., more than 15%), the student will likely need to provide further explanation on the project idea and a penalty might be imposed on the submission in case no proper justification is provided.

## **What you need to do**

**Before you begin, make sure you:**

- You are highly recommended to review the “inspiring” materials provided [here](#) to select a topic that you would like to work on. Also, you are highly recommended to propose your own interesting and novel topic and please feel free to discuss it with your tutors to ensure the topic is suitable.
- Download the **marking rubric** (available on Moodle) as guidance on how you will be assessed.

**Choose a data science project topic, and then:**

1. **Do preliminary research** about your project topic and the relevant technologies
2. **Write and submit your proposal** with cited references (Assignment 1)
3. **Research and prepare your final report** with cited references, **make a recorded presentation and submit your report** (Assignment 3).

You are free to modify the initial proposal sections submitted for Assignment 1 (especially in response to feedback from your marker), or even change topics, when you are working on Assignment 3.

## **How to Submit**

Once you have completed your work, take the following steps to submit your work. Penalties may be applied to your marks if the following instructions are not followed.

1. Please finish and save the project proposal or report first using a word processing tool (e.g., Microsoft Word), then save the document in the PDF format and submit it on Moodle.
2. Please ensure you **name the file containing your proposal/report** correctly using the following format:  
*FirstName\_StudentNumber\_AssignmentNumber.pdf*  
*e.g., Guanliang\_12345678\_Assignment1.pdf or*  
*Guanliang\_12345678\_Assignment3.pdf*
3. Upload your assignment file in the corresponding assignment link provided on Moodle.

### **Those unable to attend week 12 applied class for interview:**

Those who cannot attend their original applied class and would like to attend another applied class to have the interview, please first contact the tutors of the class that you would like to attend to check whether the tutors have additional capacity to accommodate you and if yes, then you can join the class for the interview. We cannot guarantee that you will be admitted to another applied class if you cannot attend your original class due to the limited capacity the teaching team has.

### **Further advice on the assignment:**

Here is some further advice from the teaching team regarding the assignment:

1. Make sure to carefully read the assignment specification above.
2. The project should be data-centred -- ideally combining multiple sources of data to develop your own project that can solve a real-world problem.
3. The project should contain a clear statement of the problem being tackled. What is the objective/purpose of the project?
4. Ensure that the project's novelty and benefits are clearly communicated. What makes the project novel? Will it provide financial benefits or contribute to social good?
5. The report needs to be "telling a story", and to be convincing somebody to "invest in your project" so that it can be built.
6. Try not to make the project too broad. It should be an achievable data science project.
7. Read up as much as you can on the particular topic you've chosen in order to be able to describe the data (and software) requirements of the project.
8. Make it clear where the data would come from for the project:
  - o Is the data proprietary? How would it be collected?
  - o If the data is public, you should do some exploratory data analysis on it.
9. What preprocessing would be needed? How would the data need to be preprocessed before it can be used? What software might be needed? Can the preprocessing be distributed?
10. Finally, make sure you've seen the set of possible section headings suggested above and structure your report accordingly.